

Technical Document #607: Retrieval Augmented Generation - Comprehensive Overview

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Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.
- Hybrid search combines keyword-based and semantic search.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.